

Humza Hassan Khan

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Summary

Aerospace Engineering graduate from the University at Buffalo with hands-on experience in aircraft, rocket, and spacecraft design. Led AIAA Design/Build/Fly to a 12th place international finish out of 112 teams. Skilled in CFD, FEA, MATLAB, propulsion modeling, and composite and CNC manufacturing, with a proven ability to drive projects from first-principles analysis through fabrication, testing, and flight validation.

Education

University at Buffalo, B.S. Aerospace Engineering (**Magna Cum Laude**)

2021 – 2025 | Buffalo, NY

CGPA: 3.624, Dean's List Standing in Fall & Spring 2021, 2025

Projects & Engineering Experience – [Portfolio \(GitHub\)](#)

AIAA Design/Build/Fly (DBF) – Senior Student Engineer (Manufacturing & Fabrication)

2022 – 2025

- Designed and 3D-printed flight-critical components (flap hinges, control surface mounts, vertical stabilizer brackets) and fabricated vacuum-bagged carbon fiber molds using wet layup and peel-ply techniques.
- Performed CFD simulations in ANSYS Fluent and static structural FEA to optimize component geometry for minimum drag and maximum load-bearing efficiency under flight loading conditions.
- Conducted parametric airfoil studies in XFLR5 (panel method) across a range of Reynolds numbers to select optimal wing section profiles balancing CL/CD ratio and stall characteristics.
- Directed the design, fabrication, and flight testing of **8 aircraft prototypes** over 3 seasons with varied aerodynamic configurations & missions.
- Placed **12th internationally out of 112 teams** in DBF 2025, a significant improvement from 53rd place when first joined.

CNC Laser Cutting Specialist (DBF, AIAA Region 1 Student Conference)

2022 – 2025

- Reduced positional cutting tolerances from $\pm 0.2\text{mm}$ to $\pm 0.04\text{mm}$ through precision mirror alignment, focal length calibration, and feed rate optimization on a CO₂ laser system.
- Manufactured structural airframe components (fuselage frames, rib stations, motor mounts, control surface linkages) from aircraft-grade plywood and balsa sheet stock.
- Authored a comprehensive laser cutter operations and safety manual; led hands-on training workshops for 15+ new team members across two academic years.
- Designed and laser-engraved custom award plaques installed on AIAA Region 1 Student Conference trophies.

Senior Design Project – Hermes UAV (Wingtip Propulsion Study)

2024 – 2025

- Designed and fabricated a 7 lb, 5 ft wingspan fixed-wing UAV with interchangeable motor mounting architecture to experimentally compare wingtip-mounted vs. mid-wing propulsion effects on induced drag and handling qualities.
- Constructed airframe using CNC hotwire-cut EPS foam cores with fiberglass/epoxy wet layup skins, 3D-printed control surface hinges and motor nacelles, and laser-cut plywood bulkheads; applied GD&T principles for interchangeability of modular mounts.
- Instrumented UAV with an onboard flight data recorder sampling airspeed, control surface deflections, and IMU data at 100 Hz; post-processed datasets in MATLAB to extract performance metrics and comparative time-series plots between configurations.
- Achieved a peak recorded airspeed of **~107 mph**; MATLAB analysis confirmed consistent longitudinal stability margins, validating that propulsion placement had negligible effect on trim characteristics at cruise.

Engineering Leadership

UB AIAA Student Chapter – Vice President & Lab Safety Officer

2024 – 2025

- Elected Vice President of a 60+ member organization; jointly managed an annual operating budget of **~\$45,000**, overseeing procurement, travel, and equipment expenditures.
- Organized logistics for national competition trips, club meetings, and AIAA professional chapter engagements; maintained communication with industry sponsors and alumni mentors.
- Led club-wide recruitment drives and onboarding; coordinated sub-team leads across DBF, Micro-G NExT, and Battlebots.
- Drafted and enforced lab safety protocols and equipment SOPs for all shop tools and work areas.

UB School of Engineering & Applied Sciences – Undergraduate Teaching Assistant

Spring 2025

- Graded lab quizzes and reports for MAE385: Engineering Materials Lab; instructed students during experiments covering tensile testing, hardness measurement, and materials characterization.
- Ensured experimental data integrity by cross-checking student results against theoretical values; enforced lab safety protocols for 40+ students.

UB AIAA Student Chapter – Events Coordinator

2023 – 2024

- Coordinated logistics, scheduling, and attendance for all sub-teams (DBF, Micro-G NExT, Battlebots) and club-wide events throughout the academic year.
- Managed club social media (Instagram, Discord) and drafted sponsorship proposals to secure external funding; organized volunteering events with the Students Association to secure additional funding (**~\$6,000**).

Technical Skills

Design & Simulation

SolidWorks, Autodesk Fusion 360, ANSYS Fluent, ANSYS Static Structural, XFLR5, OpenRocket, Trocen Laser CAD, Grbl, WingWire.

3D Printing

Cura, Bambu Studio, Orca Slicer; LW-PLA, HTPLA, Brass-fill, PLA/PETG; FDM process optimization.

Programming

MATLAB, Python, C++, Visual Basic, HTML.

Manufacturing

CNC laser cutting, CNC hotwire cutting, vacuum-bag composite layups, precision lathe, GD&T, soldering, sanding, finishing, polishing

Languages (Fluent)

English, Urdu, Punjabi, Hindi